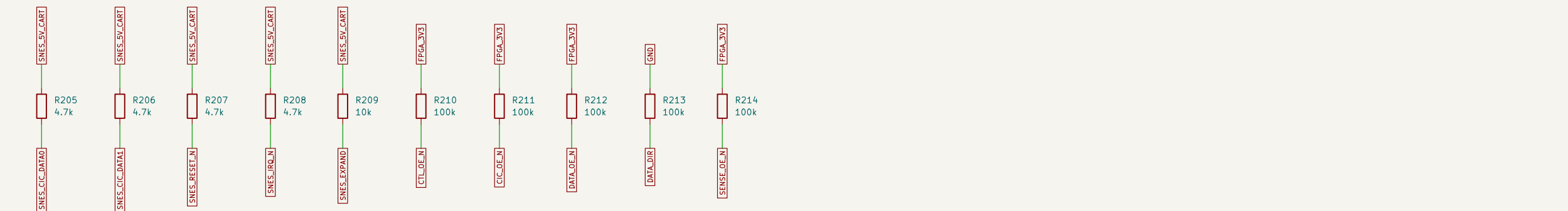
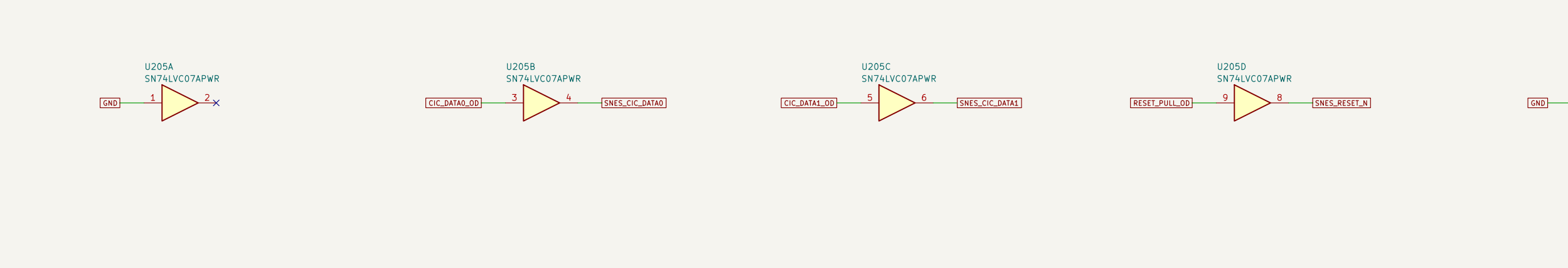
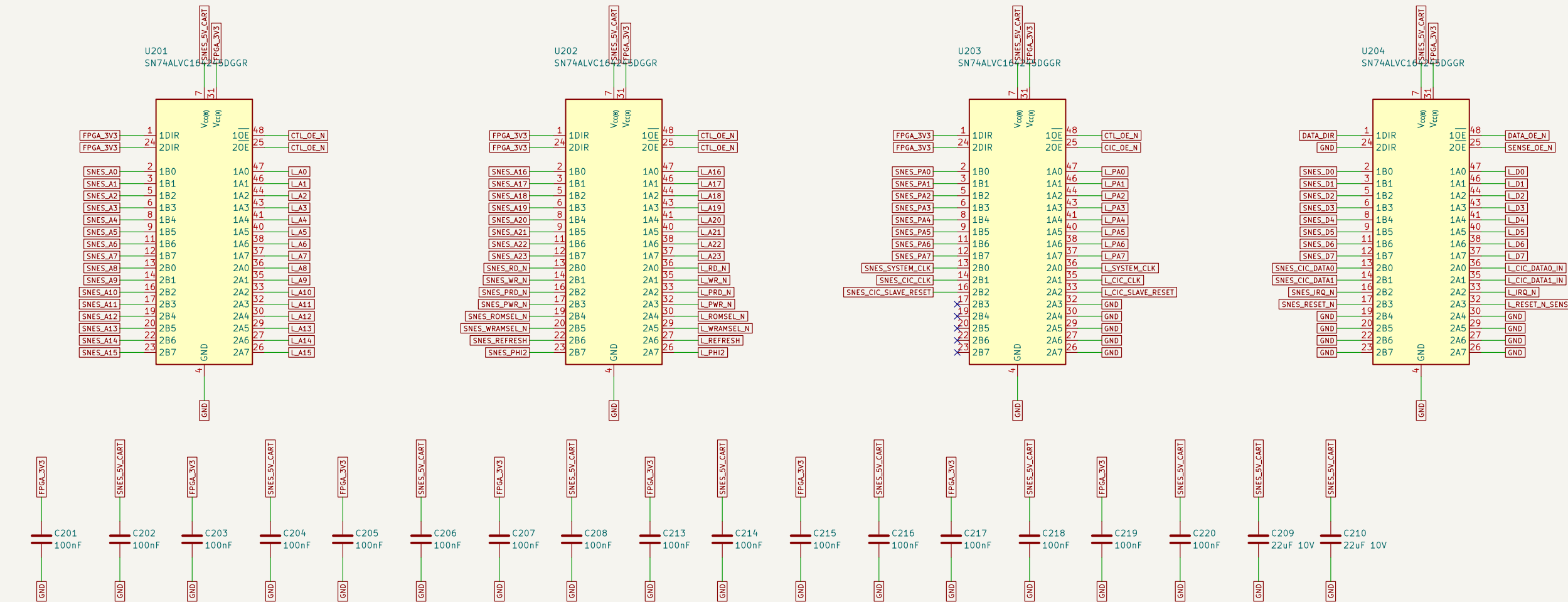
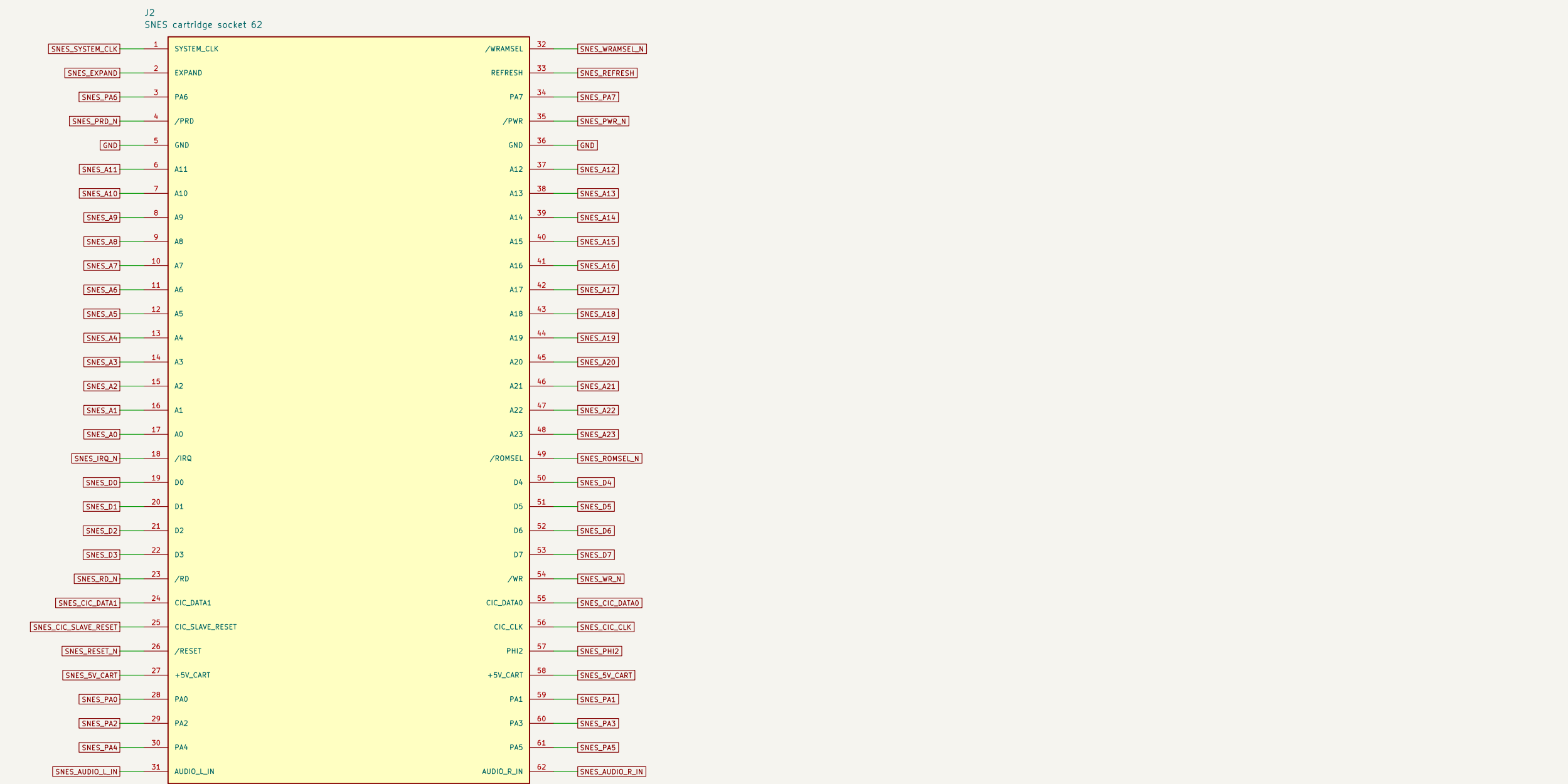




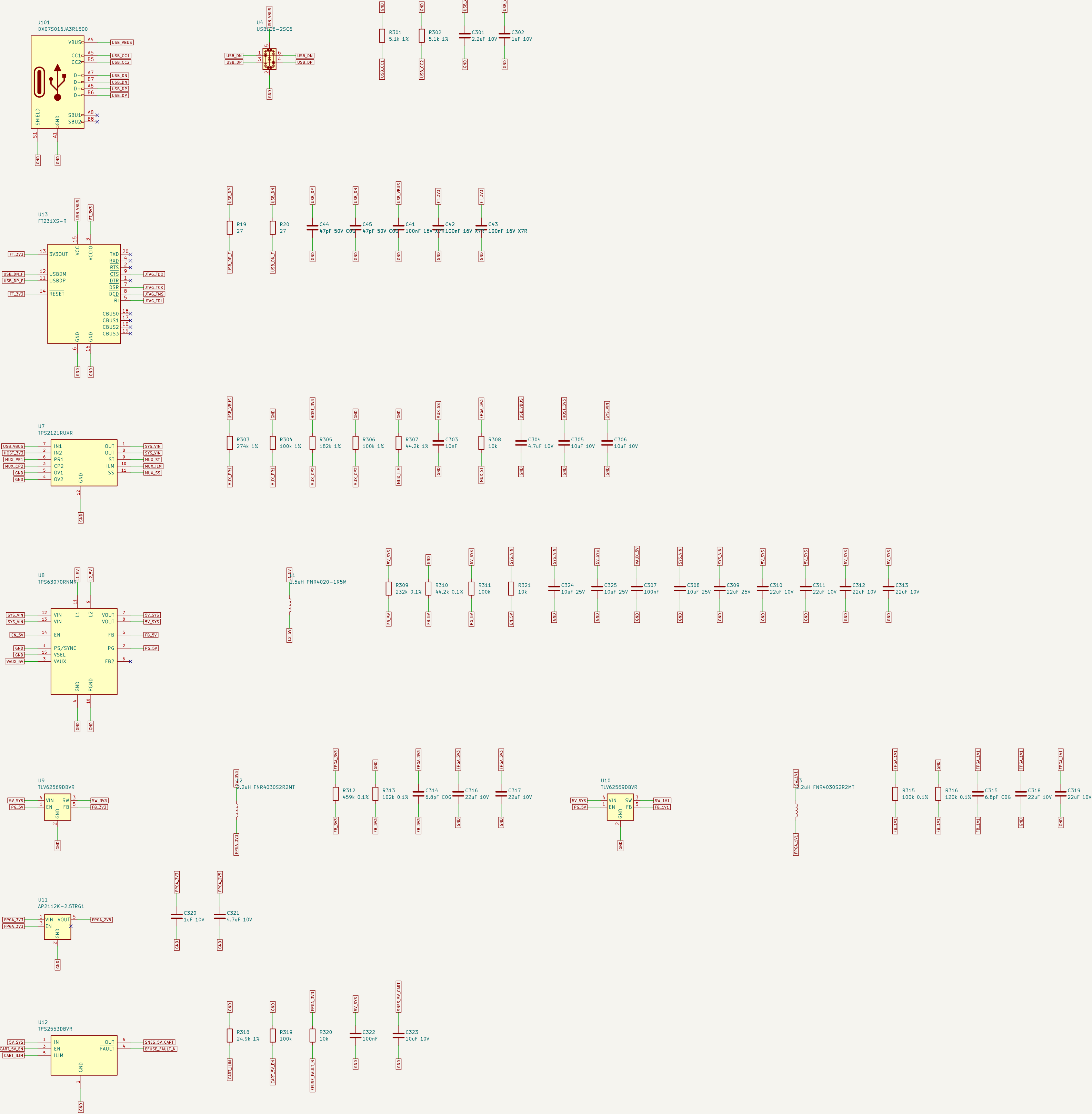
SN64 v2 FPGA SHEET - LFESU-B5F CABGA381 (-BBG381, Mouser stock), configuration flash, 27 MHz clock, JTAG, telemetry ADC, reset supervisor.
Pin map: [interfaces/fpga-pin-map.csv](#) (tools/pin_plan.py); every signal on rings 1-3; constraints [fpga/constraints/sn64_board.lpf](#).



SN64 v2 CARTRIDGE SHEET – N64 edge J1 and SNES socket J2 on the one board, four SN74ALVC164245 translators (A = 3.3 V, B = 5 V), SN74LVC07A open–drain driver for the 5 V lines the FPGA pulls.



SN64 v2 POWER SHEET – TPS2121 input USB (USB priority) –> TP563070 buck-boost 5V_SYS –> TLV62569 x2 (3V3, 1V1), AP2112K–2.5V LDO (2V5), TPS2553 cartridge switch, USB–C receptacle with the loader chip (FT231X on the FPGA JTAG port). Converter values are the v1 ones (verified pin tables).



DESIGN DRAFT – NOT FOR MANUFACTURE

Sheet: /Power and USB/
File: power.kicad_sch
Title: SN64 v2 – Power and USB
Size: A1 Date: 2026–09–30 Rev: v2 draft 0.1
Kicad E.D.A. 10.0.6 Id: 4/4